Prakash Chourasia

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Georgia State University
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RESEARCH INTERESTS

- Machine Learning, Artificial Intelligence, Data Mining, Federated Learning, Representation learning, Electronic Health Records, Clinical Data, Kernel methods, Algorithms, Bioinformatics, and Drug Discovery.

EDUCATION

- Georgia State University, Atlanta — Fall 2021 - Present
  Ph.D (Computer Science) , **Advisor:** [Dr. Murray Patterson] — GPA:3.86
- Georgia State University, Atlanta — Fall 2014 - Spring 2016
  Master of Science (Computer Science), **Advisor**: [Dr. Ying Zhu] — GPA:3.54
- Indira Gandhi National Open University New Delhi — June 2011 - May 2013
  Post Graduate Diploma (Information Security) — Percentage: 70.25
- Rajiv Gandhi Technical University Bhopal — August 2007 - June 2011
  Bachelor of Engineering (Information Technology), **Advisor**: [Lucky Sharma] — Percentage: 70.66

WORK EXPERIENCE

- Johnson & Johnson (J&J) - Atlanta USA — Data Science Research Intern — **May 2024 - Aug 2024**
- Interactive Communication (InComm) - Atlanta USA — Software Engineer-II — **Oct 2019 - Aug 2021**
- Amerisave Mortgage Corporation - Atlanta USA — Software Developer — **June 2016 - Oct 2019**
- Tata Consultancy Services - India — System Engineer — **Jan 2012 - June 2014**

PUBLICATIONS

- *Conference Papers (27)*

27 Somiya Rauf, Hafsa Farooq, **Prakash Chourasia**."Machine Learning Approaches for Radiotherapy in Head and Neck Cancer" PAKDD 2026 (Under Review)

26 **Prakash Chourasia**, Sarwan Ali and Murray Patterson."Evaluating δ-Hyperbolicity of LLM Embeddings for Personalized Health Recommendations" IEEE Bigdata 2025 (Under Review)

25 **Prakash Chourasia**, Sajjad Fouladvand."Patient Real-World Data Representation and Clustering using Large Language Models" ICLR 2025 (Under Review)

24 Sarwan Ali, Haris Mansoor, **Prakash Chourasia**, and Murray Patterson."Computing Gram Matrix for SMILES Strings using RDKFingerprint and Sinkhorn-Knopp Algorithm" ISBRA 2025 (Under Review)

23 Taslim Murad, **Prakash Chourasia**, Sarwan Ali and Murray Patterson."Neuromorphic Spiking Neural Network Based Classification of COVID-19 Spike Sequences" ISBRA 2025

22 Sarwan Ali, Tamkanat E Ali, **Prakash Chourasia**, and Murray Patterson. "Position specific scoring is all you need? Revisiting protein sequence classification tasks" RECOMB-CG 2025, "A rank conference". [PDF]

21 Sarwan Ali, **Prakash Chourasia**, Haris Mansoor, and Murray Patterson. "MIK: Modified Isolation Kernel for Biological Sequence Visualization, Classification, and Clustering" ML4H 2024, "A rank conference". [PDF]

20 **Prakash Chourasia**, Heramb Lonkar, Sarwan Ali, and Murray Patterson."EPIC: Enhancing Privacy through Iterative Collaboration" SIMBig 2024, [PDF]

19 **Prakash Chourasia**, Tamkanat E Ali, Sarwan Ali, and Murray Patterson."DWFL: Enhancing Federated Learning through Dynamic Weights" SIMBig 2024, [PDF]

18 Sarwan Ali, Haris Mansoor, **Prakash Chourasia**, Imdad Ullah Khan, and Murray Patterson."Preserving Hidden Hierarchical Structures: Poincaré Distance for Enhanced Genomic Sequence Analysis" SIMBig 2024, [PDF]

17 Sarwan Ali, **Prakash Chourasia**, and Murray Patterson. "DeepPWM-BindingNet: Unleashing Binding Prediction with Combined Sequence and PWM Features" ICONIP 2024,
"A rank conference". [PDF]

16 Taslim Murad, **Prakash Chourasia**, Sarwan Ali, and Murray Patterson."Advancing Protein-DNA Binding Site Prediction" ICONIP 2024,
"A rank conference". [PDF]

15 Sarwan Ali, Haris Mansoor, **Prakash Chourasia**, Yasir Ali, and Murray Patterson."Gaussian Beltrami-Klein Model for Protein Sequence Classification: A Hyperbolic Approach" ISBRA 2024

14 Sarwan Ali, Tamkanat E Ali, **Prakash Chourasia**, and Murray Patterson."A Universal Non-Parametric Approach For Improved Molecular Sequence Analysis" PAKDD 2024 (preprint),
"A rank conference", [PDF]

13 Zahra Tayebi, Sarwan Ali, **Prakash Chourasia**, Taslim Murad, and Murray Patterson. "T Cell Receptor Protein Sequences and Sparse Coding: A Novel Approach to Cancer Classification." ICONIP (2023), [PDF]
"A rank conference".

12 **Prakash Chourasia**, Taslim Murad, Zahra Tayebi, Sarwan Ali, and Murray Patterson. "Efficient classification of sars-cov-2 spike sequences using federated learning" The International Conference on Information Management and Big Data (SIMBiG) 2023, [PDF]

11 Sarwan Ali, **Prakash Chourasia**, and Murray Patterson."Expanding Chemical Representation with k-mers and Fragment-based Fingerprints for Molecular Fingerprinting" The International Conference on Information Management and Big Data (SIMBiG) 2023, [PDF]

10 **Prakash Chourasia**, Taslim Murad, Sarwan Ali, and Murray Patterson. "Enhancing t-SNE Performance for Biological Sequencing Data through Kernel Selection." International Symposium on Bioinformatics Research and Applications, (ISBRA) 2023. [PDF]

9 Sarwan Ali, **Prakash Chourasia** and Murray Patterson. "PDB2Vec: Using 3D Structural Information For Improved Protein Analysis" International Symposium on Bioinformatics Research and Applications, (ISBRA) 2023. [PDF]

8 Sarwan Ali, Haris Mansoor, **Prakash Chourasia** and Murray Patterson. "Hist2Vec: Kernel-Based Embeddings for Biological Sequence Classification." International Symposium on Bioinformatics Research and Applications, (ISBRA) 2023. [PDF]

7 **Prakash Chourasia**, Zahra Tayebi, Sarwan Ali, and Murray Patterson. "Empowering Pandemic Response with Federated Learning for Protein Sequence Data Analysis." In 2023 International Joint Conference on Neural Networks (IJCNN), pp. 01-08. IEEE, 2023, "A rank conference" [PDF][Slides].

6 Sarwan Ali, **Prakash Chourasia**, and Murray Patterson. "When Biology has Chemistry: Solubility And Drug Subcategory Prediction using SMILES Strings." International Conference on Learning Representation (ICLR), 2023, "A* rank conference" [PDF].

5 Ali, Sarwan, Babatunde Bello, **Prakash Chourasia**, Ria Thazhe Punathil, Pin-Yu Chen, Imdad Ullah Khan, and Murray Patterson. "Virus2Vec: Viral Sequence Classification Using Machine Learning." Conference on Health, Inference, and Learning (CHIL 2023), "Selected for Oral presentation (12% acceptance rate)" [PDF][Slides].

4 Taslim Murad*, **Prakash Chourasia***, Sarwan Ali*, and Murray Patterson. "Hashing2vec: Fast embedding generation for sars-cov-2 spike sequence classification." In Asian Conference on Machine Learning (ACML 2023), pp. 754-769, 2023. (32% acceptance rate) [PDF]. * **Equal Contribution**

3 **Prakash Chourasia**, Sarwan Ali, Murray Patterson."Effect of Informative Initialization on the Quality of t-SNE For Biological Sequences" IEEE BigData 2022.
(18.6% acceptance rate) [PDF] [Slides].

2 Sarwan Ali, Taslim Murad, **Prakash Chourasia** and Murray Patterson. "Spike2Signal: Classifying Coronavirus Spike Sequences with Deep Learning". In the IEEE Eight International Conference on Big Data Computing Service and Applications (Big Data Service) 2022.

1 **Prakash Chourasia**, Sarwan Ali, Simone Ciccolella, Gianluca Della Vedova, and Murray Patterson. " Clustering SARS-CoV-2 variants from raw high-throughput sequencing read data". In the 11th International Conference on Computational Advances in Bio and Medical Sciences, ICCABS (2021).

- *Journals (10)*

10 Taslim Murad, **Prakash Chourasia**, Sarwan Ali, and Murray Patterson."DANCE: Deep Learning-Assisted Analysis of Protein Sequences Using Chaos Enhanced Kaleidoscopic Images" JCB 2025 (Under Review)

9 Sarwan Ali, Tamkanat E Ali, **Prakash Chourasia**, and Murray Patterson. "Compression and k-mer based Approach For Anticancer Peptide Analysis" TCBB (Under Review)

8 Sarwan Ali, **Prakash Chourasia**, Bipin Koirala, and Murray Patterson. "Approximate CCP-Based Protein Sequence Analysis" TCBB 2025
Impact Factor: 3.6 (2023)

7 Sarwan Ali, Tamkanat E Ali, Haris Mansoor, **Prakash Chourasia**, and Murray Patterson. "Hist2Vec: A histogram and kernel-based embedding method for molecular sequence analysis" Expert Systems with Applications (2025): 126859.
Impact Factor: 7.5 [PDF].

6 Sarwan Ali, Madiha Shabbir, Haris Mansoor, **Prakash Chourasia** and Murray Patterson."Elliptic Geometry-Based Kernel Matrix for Improved Biological Sequence Classification" Knowledge-Based Systems Volume 304, Nov 2024,
Impact Factor: 7.2 [PDF]

5 Sarwan Ali, **Prakash Chourasia**, and Murray Patterson."When Protein Structure Embedding Meets Large Language Models" MDPI Genes 2024
Impact Factor: 3.5. [PDF].

4 Sarwan Ali, **Prakash Chourasia**, and Murray Patterson."From PDB Files to Protein Features: A Comparative Analysis of PDB Bind and STCRDAB datasets" Medical & Biological Engineering & Computing 2024
Impact Factor: 3.2 [PDF].

3 **Prakash Chourasia**, Sarwan Ali, Simone Ciccolella, Gianluca Della Vedova, and Murray Patterson. "Reads2vec: Efficient embedding of raw high-throughput sequencing reads data." Journal of Computational Biology 30, no. 4 (2023): 469-491
Impact Factor: 1.7 [PDF].

2 Sarwan Ali, **Prakash Chourasia**, Zahra Tayebi, Babatunde Bello, and Murray Patterson. "ViralVectors: Compact and Scalable Alignment-free Virome Feature Generation" Medical & Biological Engineering & Computing 2023
Impact Factor: 3.2 [PDF].

1 Sarwan Ali, Babatunde Bello, **Prakash Chourasia**, Ria Thazhe Punathil, Yijing Zhou, and Murray Patterson. "PWM2Vec: An Efficient Embedding Approach for Viral Host Specification from Coronavirus Spike Sequences." MDPI Biology 2022.
Impact Factor: 5.07 [PDF].

HONOURS AND AWARDS

- Outstanding Graduate Student Award **2025**
- Fraser Travel Award **2025**
- **Best Paper Awadrd** SIMBig 2023 **2023**
- Molecular Basis of Disease (MBD) **Ph.D. Fellowship (funding for Ph.D.)** **2023**
  Georgia State University, Atlanta, GA, USA
- Student Travel Award [ICDM 2023], [IJCNN 2023], [ICLR 2023]) **2023**
- Home Depot HACKATHON Winning team - 3rd Prize **2016**
- President Award for Boy Scouts - Eagle Scout
  (By - Dr. APJ Abdul Kalam, Former President of India) **2007**

TEACHING AND ACADEMIC EXPERIENCE

SERVICES

- Organizing committee of the XAI Section (WCCI 2024)
- Reviewer - Conference/Journal (IJCNN 2024, ACML 2023, ISBRA 2023, MBEC)

TEACHING EXPERIENCE

- Developed video-based instructional content on YouTube to reach a broader audience. YouTube
- Georgia State University, Atlanta, USA
  - ***Teaching Fellow (Instructor)***
    * CSC 1302 Principles of Computer Science II **Spring 2022**
      Worked as instructor for a class of 150 students
  - ***Graduate Teaching Assistant/ Lab Instructor***
    * CSC 4850/6850 Machine Learning [Instructor: [Md. Mahfuzur Rahman]] **Summer 2025**
    * CSC 4352 Capstone II-CTW [Instructor: [William Gregory Johnson]] **Spring 2025**
    * CSC 1301 Principles of Computer Science I [Instructor: [Hossein Saghaeiannejad Esfahani]] **Spring, Fall 2024**
    * CSC 4370 Web Programming [Instructor: [Louis Henry]] **Spring, Summer, Fall 2023**
    * CSC 3320 System-Level Programming [Instructor: [Michael Weeks]] **Fall 2022**
    * CSC 3210 Computer Org and Programming [Instructor: [Xucan Chen]] **Fall 2021**

COURSES

- Advance Machine Learning
- Advance Deep Learning
- Data Mining
- Computational Epidemiology
- Advance Algo in Bioinformatics
- Bio-Statistics

TECHNICAL SKILLS

- **DATA TOOLS:** D3.JS, Google Charts, MATLAB, R-Programming
- **PROGRAMMING SKILLS:** Python, Java, C, C++, Android, Python, Lucee, ColdFusion, Spring, Node.js
- **TOOLS AND IDE:** Microsoft visual studio, Eclipse, NetBeans, STS, Android Studio, Django, and BIDS
- **DATABASE:** BI, SSIS, SSRS, MS SQL Server 2012-14, Oracle PL/SQL
- **WEB TECHNOLOGY:** Bootstrap, HTML, CSS, Kendo UI, JavaScript, AJAX, JQuery. Reactive JS, React JS